

Sofia Sarak

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EDUCATION

Master of Environmental Data Science (June 2026)

Bren School of Environmental Science & Management, University of California, Santa Barbara (UCSB)

Coursework: [Geospatial Analysis](#), [Data Visualization & Communication](#), Machine Learning, [Data Management](#), Statistics

Bachelor of Arts in Earth & Environmental Sciences, Economics (May 2025)

Wesleyan University, Middletown, CT

Coursework: Econometrics, Environmental Econ, Labor Econ, Education Econ, Renewable Energy, Soils, Sedimentology

Study Abroad: University of Otago, New Zealand (Fall 2024); Universidad San Francisco de Quito, Galápagos Islands (1/24)

PROJECT EXPERIENCE

Power Lines and People: Mapping How Distribution Grid Constraints Shape Resilient and Equitable Energy Transition

Master's Capstone Project, UCSB (12/25—6/26)

- Developed a [reproducible analytical framework](#) in Python and QGIS that spatially links 11.5 million California homes to electric distribution grid and sociodemographic data, creating a previously non-existent geospatial data package
- Identified communities vulnerable to grid capacity constraints using regression and random forest models
- Visualized and interpreted cleaned data; synthesized findings in technical documentation and public presentation

Ants, Plants, and Isotopes: Investigating the Trophic Levels of the Guánica Dry Forest

Earth & Environmental Sciences Senior Capstone Project, Wesleyan University (9/24—3/25)

- Determined trophic level of 5 different ant species through isotope analysis, using methods from literature
- Led group's data analysis efforts, using R to explore differences between ecological niches in Puerto Rico

PROFESSIONAL EXPERIENCE

Career Development Student Assistant, **Bren School of Environmental Science & Management** (10/25—Present)

- Collect, summarize, and visualize environmental data job market data using R and assemble in [quarterly reports](#)
- Develop marketing material such as an email drip campaign, alumni outcome analyses, and program one-pagers
- Update the [Bren Student Data Explorer](#) (R Shiny Dashboard) with the most recent alumni placement data

Conservation Intern, **Westport Conservation Department**, Westport, CT (07/26—08/26)

- Develop an interactive geospatial data viewer for urban water quality data using reproducible workflows in R
- Wrangle and aggregate decades of sampling data to be comprehensible for the general public
- Work with conservation director and IT staff to launch viewer on the Conservation Department's website

Lab Intern, **Chernoff Lab (Bailey College of the Environment) – Wesleyan University** (9/23—5/25)

- Performed morphometric analysis on 1,000+ freshwater minnow samples using R and ImageJ software
- Organized and managed data files representing 15 different geographic regions and populations
- Provided feedback on visuals and thesis presentations during weekly lab meetings

Field and Lab Intern, **Harbor Watch at Earthplace**, Westport, CT (5/23—8/23)

- Collected, processed, and determined bacteria concentrations of water samples across Fairfield County, CT
- Performed data entry on bacteria concentration and marine species and created pivot tables in MS Excel
- Compiled reports regarding research findings and potential projects, such as location scouting

SKILLS AND ADDITIONAL EXPERIENCE

Data Science: R/RStudio (tidyverse, ggplot2, terra, sf, stars, plotly, leaflet), Python (numpy, pandas, geopandas, matplotlib, xarray, scikit-learn), GitHub, SQL, Stata, econometric analysis, geospatial analysis, vector-based graphic design, R Shiny apps

Languages: Russian (fluent), Ukrainian (intermediate), Spanish (intermediate)

Awards: UCSB Graduate Division Public Speaking Certificate (3/26); MEDS Academic Achievement Award (6/26)

Additional Experience: *Course Assistant*, Economics Dept. – Wesleyan University; *Dean's Peer Tutor*, Wesleyan University